

Telecom Customer Churn Analysis

1. Project Overview and Objective

This project analyzes customer churn in a telecom company to identify the key factors that influence customers to discontinue their services. The analysis involves data cleaning, transformation, exploratory data analysis (EDA) in Excel, and interactive dashboard creation in Power BI.

A telecom company is experiencing customer attrition (churn), which leads to reduced revenue and increased customer acquisition costs. The company needs to identify the key factors influencing customer churn, such as customer demographics, service usage, contract type, payment method, tenure, monthly charges, referrals, and customer support services.

The goal is to analyze churn behavior, discover the key drivers of customer attrition, and provide actionable insights that help the business improve customer satisfaction, increase retention, and reduce future churn.

2. Data Sources

- **Source Description and Timeline:** <https://mavenanalytics.io/data-playground/telecom-customer-churn> and At the end of fiscal quarter (June-2022).
- **Domain:** Sales Analytics

3. Problem Statement

- Analyze the key factors contributing to increasing customer churn, which negatively impacts customer retention and revenue, driven by rising competition, evolving customer preferences, and service-related issues.
 - Identify the customer segments, including different age groups, contract types, and tenure groups, that exhibit the highest customer churn rates to support targeted retention strategies.
 - Examine the influence of referrals and marketing offers on customer retention.
 - Evaluate the impact of phone service usage, internet service usage, and internet service type on customer churn.
 - Assess customer subscriptions to online security, online backup, streaming TV, and streaming movies to determine their effect on customer churn.
 - Investigate how monthly charges and billing patterns influence customer churn.
 - Determine the impact of population size on customer churn.
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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Customer_ID	String	Unique identifier assigned to each customer
Gender	String (Text)	Customer's gender (Male or Female)
Age	Numeric (Integer)	Age of the customer in years
Married	String (Text)	Indicates whether the customer is married (Yes/No)
Number of Dependents	Numeric (Integer)	Number of dependents associated with the customer
City	String (Text)	City where the customer resides
Zip Code	Numeric (Integer)	ZIP/postal code of the customer's primary residence
Latitude	Numeric (Decimal)	Geographic latitude of the customer's location
Longitude	Numeric (Decimal)	Geographic longitude of the customer's location
Number of Referrals	Numeric (Integer)	Number of new customers referred by the customer
Tenure in Months	Numeric (Integer)	Number of months the customer has been with the company
Offer	Categorical (Text)	Promotional offer or plan accepted by the customer
Phone Service	String (Text)	Indicates whether the customer subscribes to phone service (Yes/No)
Avg Monthly Long Distance Charges	Numeric (Decimal)	Average monthly charges incurred for long-distance calls
Multiple Lines	String (Text)	Indicates whether the customer has multiple phone lines (Yes/No)
Internet Service	String (Text)	Indicates whether the customer has an internet service subscription (Yes/No)

Internet Type	Categorical (Text)	Type of internet connection (e.g., DSL, Fiber Optic, Cable)
Avg Monthly GB Download	Numeric (Integer)	Average amount of internet data downloaded per month (GB)
Online Security	String (Text)	Indicates whether the customer subscribes to an online security service
Online Backup	String (Text)	Indicates whether the customer subscribes to an online backup service
Device Protection Plan	String (Text)	Indicates whether the customer has a device protection plan
Premium Tech Support	String (Text)	Indicates whether the customer subscribes to premium technical support
Streaming TV	String (Text)	Indicates whether the customer streams television services
Streaming Movies	String (Text)	Indicates whether the customer streams movies
Streaming Music	String (Text)	Indicates whether the customer streams music services
Unlimited Data	String (Text)	Indicates whether the customer has an unlimited data plan
Contract	Categorical (Text)	Type of customer contract (Month-to-Month, One Year, Two Year)
Paperless Billing	String (Text)	Indicates whether the customer uses paperless billing
Payment Method	Categorical (Text)	Customer's preferred payment method
Monthly Charge	Numeric (Decimal)	Total monthly charges for all subscribed services
Total Charges	Numeric (Decimal)	Total amount charged to the customer during their subscription period
Total Refunds	Numeric (Decimal)	Total amount refunded to the customer

Total Extra Data Charges	Numeric (Integer)	Total charges incurred for additional data usage beyond the plan limit
Total Long Distance Charges	Numeric (Decimal)	Total long-distance call charges accumulated by the customer
Total Revenue	Numeric (Decimal)	Total revenue generated from the customer after accounting for charges and refunds
Customer Status	Categorical (Text)	Current customer status (Stayed, Churned, or Joined)
Churn Category	Categorical (Text)	High-level category explaining why the customer churned
Churn Reason	String (Text)	Specific reason provided for customer churn
Churn Flag	Numeric (Integer)	Specified 1 for churned, 0 for Stayed and Joined
Population	Numeric (Integer)	Estimated total population residing within the corresponding ZIP code area
Age Group	String (Text)	Newly Grouped Age Column
Tenure Group	String (Text)	Newly Grouped Tenure Column
Monthly Charge Group	String (Text)	Newly Grouped Monthly charge Column
Population Category	String (Text)	Newly Grouped Population Column
Contract Length	String (Text)	Newly Grouped Contract Length Column
Revenue Category	String (Text)	Newly Grouped Revenue Category Column
Referral Category	String (Text)	Newly Grouped Referral Category Column
Risk Level	String (Text)	Newly Grouped Risk Level Column

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
 - **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.
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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.

Handled Missing Values Formulas: Replacing empty cells with appropriate values

=IF([@Phone Service]="No","0",[@Avg Monthly Long Distance Charges])

=IF([@Phone Service]="No","No Phone service",[@Multiple Lines])

=IF([@Internet Service]="No","No Internet service",[@Internet Type])

=IF([@Internet Service]="No",0,S2)

=IF([@Internet Service]="No","No Internet service",[@Online Security])

=IF([@Internet Service]="No","No Internet service",[@Online Backup])

=IF([@Internet Service]="No","No Internet service",[@Device Protection Plan])

=IF([@Internet Service]="No","No Internet service",[@Premium Tech Support])

=IF([@Internet Service]="No","No Internet service",[@Streaming TV])

=IF([@Internet Service]="No","No Internet service",[@Streaming Movies])

=IF([@Internet Service]="No","No Internet service",[@Streaming Music])

=IF([@Internet Service]="No","No Internet service",[@Unlimited Data])

=IF(OR([@Customer Status]="Stayed",[@Customer Status]="Joined"),"Not Churned",[@Churn Category])

=IF(OR([@Customer Status]="Stayed",[@Customer Status]="Joined"),"Not Applicable",[@Churn Reason])

Newly Created Calculated Columns:

Population Category:

=IF(AND([@Population]>20000,[@Population]<=50000),"Medium Population",IF([@Population]<=20000,"Low Population","High Population"))

Tenure Group:

```
=IF([@[Tenure in Months]]<=12,"New Customer",IF([@[Tenure in Months]]<=36,"Early Customer",IF([@[Tenure in Months]]<=60,"Established Customer","Long term Customer")))
```

Age Group:

```
=IF([@Age]<=29,"Young Adult",IF([@Age]<=39,"Adult",IF([@Age]<=59,"Mid Age","Senior Citizen")))
```

Monthly Charge Group:

```
=IF([@[Monthly Charge]]<=40,"Low Charge",IF([@[Monthly Charge]]<=80,"Medium Charge","High Charge"))
```

Churn Flag:

```
=IF([@[Customer Status]]="Churned",1,0)
```

- **Filtering & Sorting:**

Organized data to focus on relevant records. Use the following lookup functions to create grouped columns in the dataset.

```
=VLOOKUP([@[Zip Code]],Telecom_Customer[[Zip Code]:[Longitude]],2,FALSE)
```

```
=VLOOKUP([@[Zip Code]],Telecom_Customer[[Zip Code]:[Longitude]],3,FALSE)
```

```
=INDEX(Telecom_Customer[[City]:[ZipCode]],
```

```
MATCH([@[ZipCode]],Telecom_Customer
```

```
[Zip Code],0),1)
```

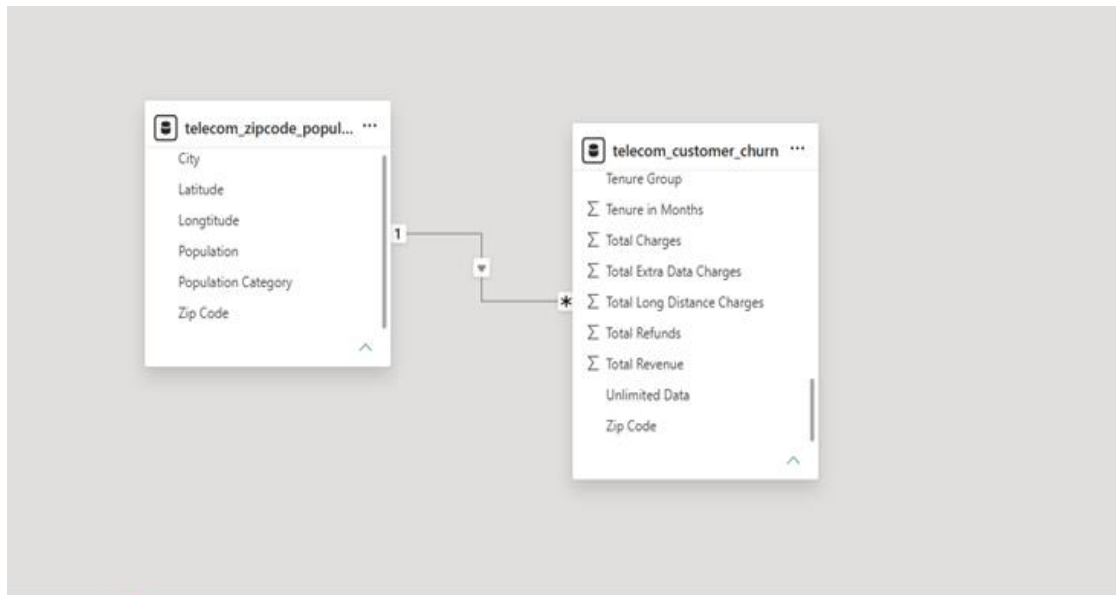
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.
- Established relationships between the fact and dimension table.

Established relation with a Zipcode between Telecom_Customer(Fact table) and Telecom_Population(Dimension table) tables.

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between telecom_customer_churn and telecom_zipcode_population tables.

Cardinality: Many to one(*:1), Cross filter direction: Both



- **Calculated Columns:**

Referral Category

Referral Category = SWITCH(

TRUE(),

telecom_customer_churn[Number of Referrals]=0,"No Referral",

telecom_customer_churn[Number of Referrals]<=3,"Low Referral",

telecom_customer_churn[Number of Referrals]<=7,"Medium Referral","High Referral")

Revenue Category

Revenue Category = SWITCH(

TRUE(),

telecom_customer_churn[Total Revenue]<1000,"Low Revenue",

telecom_customer_churn[Total Revenue]<5000,"Medium Revenue","High Revenue")

Contract Length

Contract Length = SWITCH(

telecom_customer_churn[Contract],"Month-to-Month","Short Contract",

"One Year","Medium Contract",

"Two Year","Long Contract"

)

Risk Level

```
Risk Level = SWITCH(  
    TRUE(),  
    telecom_customer_churn[Contract]="Month-to-Month"  
    && telecom_customer_churn[Tenure in Months]<=12  
    && telecom_customer_churn[Monthly Charge]>=70,"High Risk",  
    telecom_customer_churn[Contract]="Month-to-Month"  
    && telecom_customer_churn[Tenure in Months]<=24,"Medium Risk","Low  
    Risk")
```

- **DAX Measures:**

Total Customers, Stayed, joined, Churned Customers:

```
Total Customers = COUNT(telecom_customer_churn[Customer ID])  
Churned Customers = CALCULATE(COUNT(telecom_customer_churn[Customer  
ID]),telecom_customer_churn[Customer Status]="Churned")  
Stayed Customers = CALCULATE(COUNT(telecom_customer_churn[Customer  
ID]),telecom_customer_churn[Customer Status]="Stayed")  
New Customers = CALCULATE(COUNT(telecom_customer_churn[Customer  
ID]),telecom_customer_churn[Customer Status]="Joined")  
Churn Rate % = DIVIDE([Churned Customers],[Total Customers],0)*100
```

Internet & Phone Service Users:

```
Internet Users = CALCULATE(COUNT(telecom_customer_churn[Customer  
ID]),telecom_customer_churn[Internet Service]="Yes")  
Phone Service Users = CALCULATE(COUNT(telecom_customer_churn[Customer  
ID]),telecom_customer_churn[Phone Service]="Yes")  
Phone&Internet Users = CALCULATE(COUNT(telecom_customer_churn[Customer  
ID]),telecom_customer_churn[Phone  
Service]="yes",telecom_customer_churn[Internet Service]="Yes")  
Phone with No Internet Service=  
CALCULATE(COUNT(telecom_customer_churn[Customer  
ID]),telecom_customer_churn[Phone  
Service]="Yes",telecom_customer_churn[Internet Service]="No")
```


Internet Service with No Phone =

```
CALCULATE(COUNT(telecom_customer_churn[Customer  
ID]),telecom_customer_churn[Internet  
Service]="Yes",telecom_customer_churn[Phone Service]="No")
```

Total Revenue, Refunds, long distance charges, Extra data charges, Population:

Total Revenue = SUM(telecom_customer_churn[Total Revenue])

Total Refunds = SUM(telecom_customer_churn[Total Refunds])

Total Long Distance Charges = SUM(telecom_customer_churn[Total Long Distance
Charges])

Total Extra Data Charges = SUM(telecom_customer_churn[Total Extra Data Charges])

Population

Total Population = SUM(telecom_zipcode_population[Population])

Average GB Download, Monthly Charge, Tenure:

Average GB Download = AVERAGE(telecom_customer_churn[Avg Monthly GB
Download])

Average Monthly Charge = AVERAGE(telecom_customer_churn[Monthly Charge])

Average Tenure = AVERAGE(telecom_customer_churn[Tenure in Months])

- **Calculated Tables:**

Customer Summary

Customer Summary =

```
SUMMARIZE(telecom_zipcode_population,telecom_zipcode_population[City],  
"Customers Count",COUNT(telecom_customer_churn[Customer ID]),  
"Revenue",SUM(telecom_customer_churn[Total Revenue]))
```

Top 10 Cities by Revenue

```
Top 10 Cities = TOPN(10,  
SUMMARIZE(telecom_zipcode_population,telecom_zipcode_population[City],  
"Revenue",SUM(telecom_customer_churn[Total Revenue])),  
[Revenue],DESC)
```

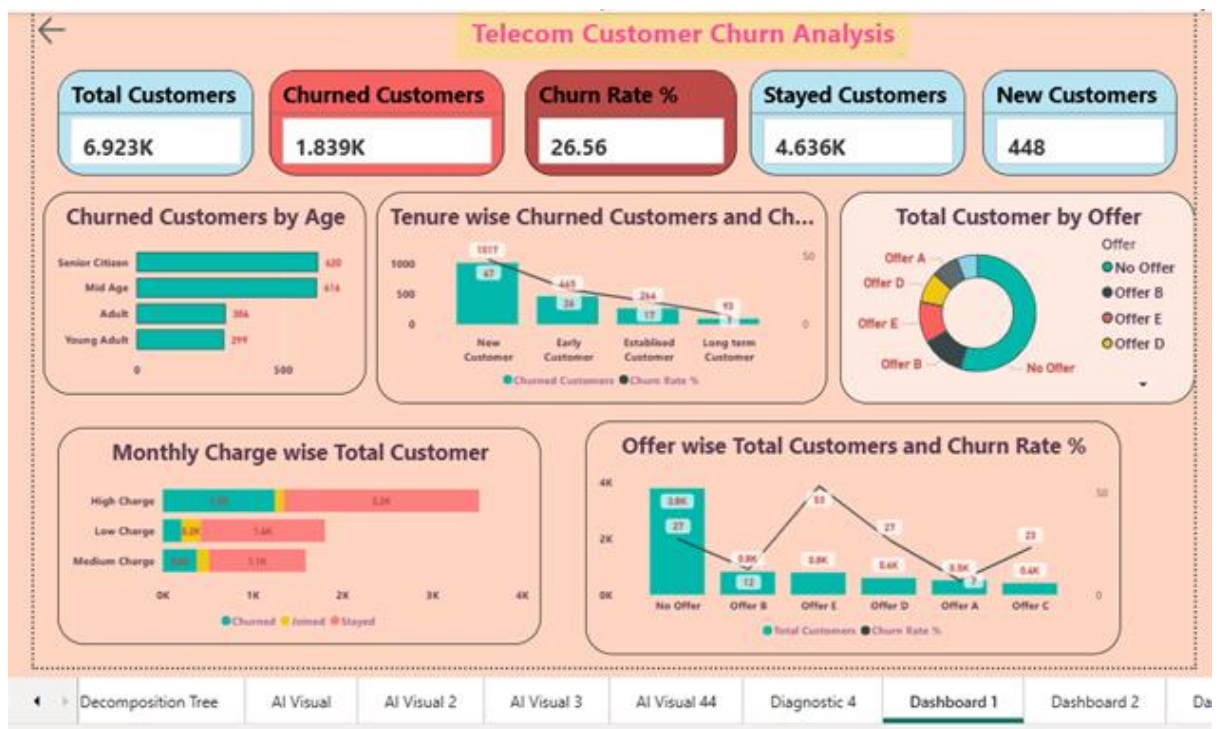
Top 5 Churn Cities

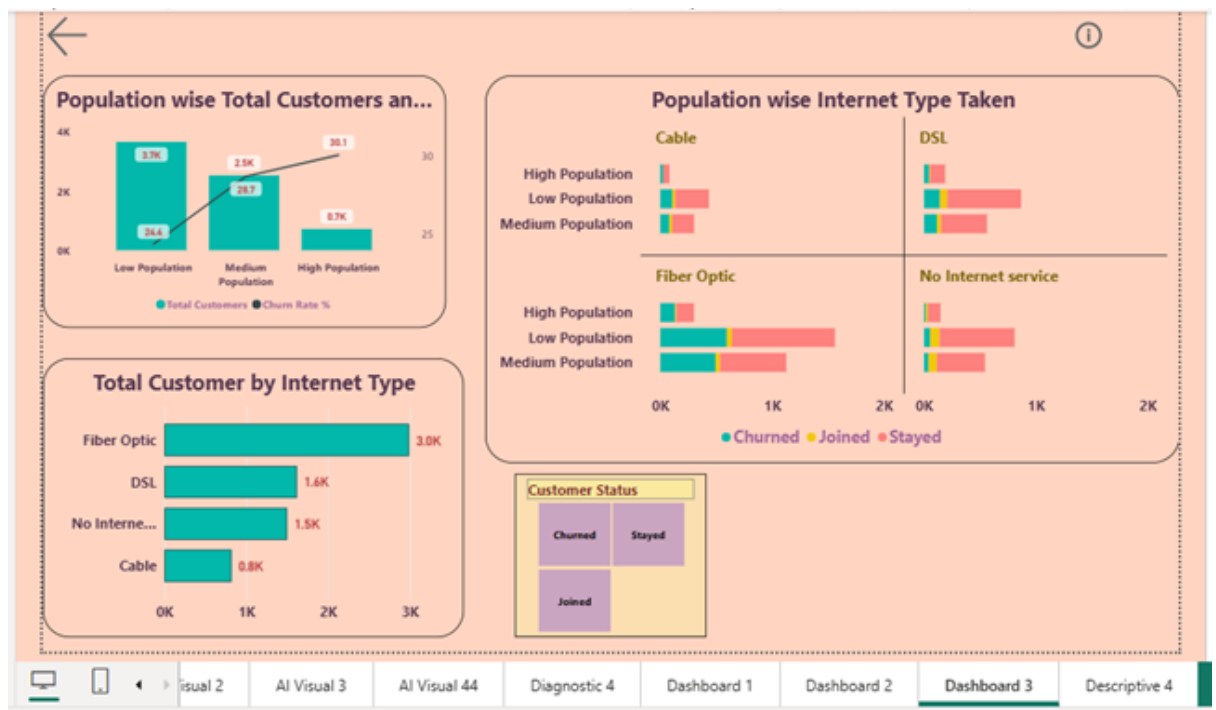
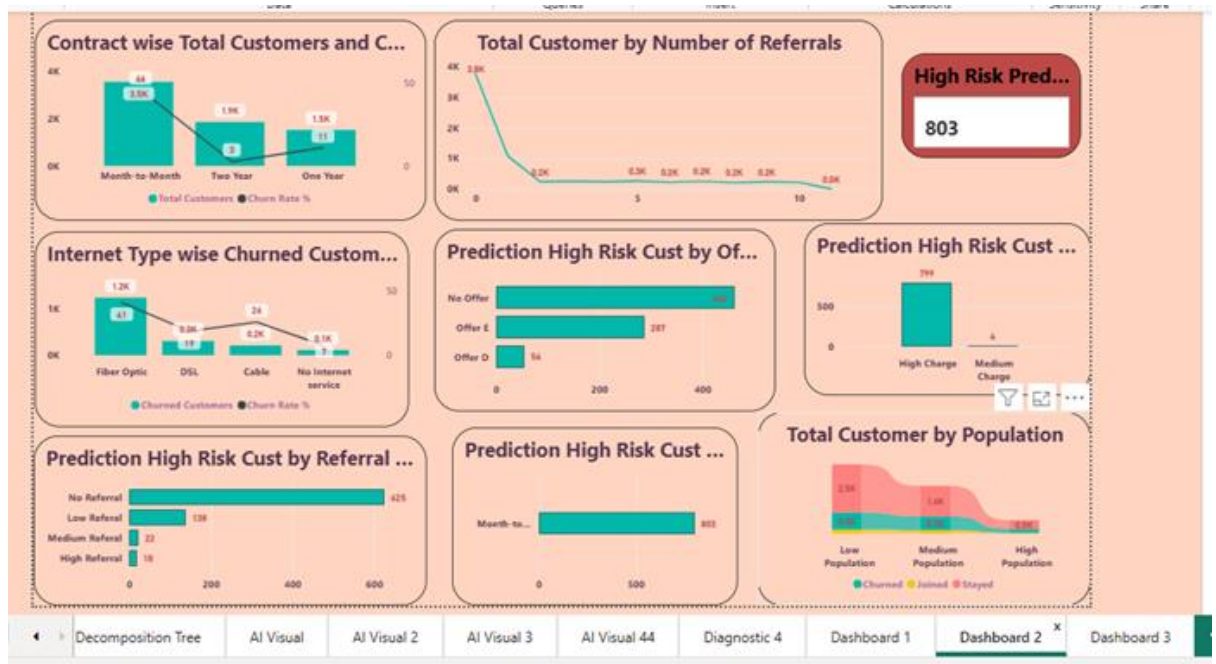
Top 5 Churn = TOPN(5,
SUMMARIZE(telecom_zipcode_population,telecom_zipcode_population[City],
"ChurnCount",CALCULATE(COUNTROWS(telecom_customer_churn),telecom_customer_churn[Customer Status]="Churned")),
[Churn Count],DESC)

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- Developed multiple visualizations (bar charts, column charts, combo charts, line charts, pie charts, scatter chart, ribbon charts, area chart, cards, and tables) based on the problem statement to communicate key insights.
- Designed an interactive Power BI dashboard using drill-downs, filters, slicers, bookmarks, and clear titles to enhance data exploration and user experience.
- Dashboard Screenshot**





9. Insights & Conclusions

Key Findings: Summarize trends, patterns, or anomalies identified in the data

Identifying key insights from the analysis

The telecom company currently serves **6,923** customers, of whom **1,839** have churned, resulting in a **26.56%** churn rate, indicating a significant customer retention challenge. Most customers are on **Month-to-Month contracts**, and this segment records the highest churn. **New customers** are the largest tenure group and experience the highest churn, while **Senior Citizens** show the highest churn among all age groups, followed closely by **Mid-**

age customers. **Fiber Optic** is the most widely used internet service and also has the highest churn compared to DSL and Cable users. Fiber Optic customers, particularly in **low-population** areas, also exhibit higher churn, mainly due to the lack of value-added services such as **Online Security, Premium Tech Support, Online Backup, Device Protection, Streaming TV, Streaming Movies, and Streaming Music**. Customers with **high monthly charges, no marketing offers, and no referrals** are more likely to churn, suggesting that pricing and customer engagement play a major role in customer loss. Although **cities with larger customer bases** report a higher number of churned customers, Looking ahead, predictive analysis identifies around **803** customers as high risk of future churn, with most of them being **new customers on Month-to-Month contracts, paying high monthly charges, receiving no marketing offers, and having no referrals**. To improve customer retention, the company should encourage customers to switch to **One-Year or Two-Year contracts**, provide targeted **retention offers** before renewal, offering **personalized discounts**, strengthen **loyalty and referral programs**, review **high pricing** by introducing better-value service bundles, and improve the quality and support of **Fiber Optic** services to reduce future churn.

Population analysis indicates that **Low Population** areas account for the highest number of churned customers (895) because they have the largest customer base. However, **Medium Population** areas record the second highest churned customers (728), indicating that customers in medium-population areas are relatively more likely to churn. **High Population** areas have the fewest churned customers (216) due to their smaller customer base, but they still exhibit a relatively high churn rate of 30.1%, indicating that customer retention remains a concern. These findings suggest that the company should focus on reducing the overall volume of churn in **Low Population** areas while implementing targeted retention strategies in **Medium and High Population areas**, where customers are more likely to churn.

10. Conclusions

The analysis indicates that customer churn is primarily driven by strong competition, better service offerings, attractive pricing, and enhanced value provided by competitors. Customer feedback reveals that many customers switched because competing providers offered better overall service quality and more appealing offers. To address this challenge, the company should focus on strengthening customer retention by promoting long-term contracts, improving Fiber Optic service quality, expanding value-added services, introducing personalized retention and loyalty programs, and offering competitive pricing and targeted marketing

campaigns. Implementing these strategies will help reduce customer churn, improve customer loyalty, increase customer lifetime value, and drive sustainable long-term revenue growth.

Because customer retention is an important challenge, and the company needs effective strategies to reduce churn and keep customers loyal. This is a significant business concern because retaining existing customers is usually more cost-effective than acquiring new ones.